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Occupational Safety & Health

Soldering Safety

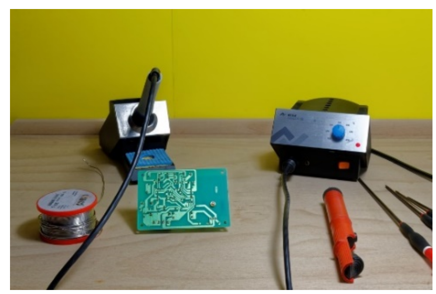
Soldering is widely used in research and industrial settings for fabricating electronic components. The process inherently poses hazards, so it is essential that best health and safety practices be implemented.

Soldering Hazards:

- **Heat & Electrical:** The soldering iron will be hot, which may cause burn injuries if contacted directly. Flammable materials may catch fire from the heat, and electric shock hazard from exposed wiring could cause injury or start a fire.
- **Rosin:** Solder used for hand soldering electronics typically contains a rosin core. While soldering, rosin generates fume that can cause eye and respiratory irritation. Repeated exposure can cause asthma.
- **Lead:** Although lead-free solder is readily available, some solders still contain lead as a major component. Hand contact with lead solder or surfaces contaminated with lead solder pose an ingestion hazard. Lead can affect the nervous system, reproductive system, kidneys, and other organs of the body.

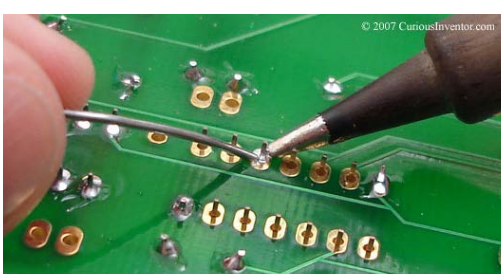
General Safe Practices:

1. Do not permit food or drinks in soldering areas.
2. Wear approved safety glasses and appropriate street clothing, including pants, closed-toe shoes, and long sleeves. Clothing made of natural fibers, including cotton, are preferable.
3. Review the Safety Data Sheet (SDS) for the solder you are using for safety information.
4. Substitute lead-containing solder with lead-free solder materials.
5. Inspect soldering equipment for damage before each use.
6. Solder in an area free of flammable materials.
7. Avoid contact with the soldering iron tip. Use the soldering iron stand.
8. Solder in a fume hood or use other local exhaust ventilation. This is especially important for operations lasting more than a few minutes. When using local exhaust ventilation:
    a. Venting to the outside is preferable.
    b. If using a system that exhausts into the room, the unit should be equipped with HEPA filtration to remove fume (condensed particulate) and chemical filters to remove vapors. Contact EH&S before setup.
9. Always wash hands with soap and water after stepping away from soldering work.



Soldering Cleanup and Waste:

All waste and residues from soldering shall be treated as hazardous waste.



1. Create a waste tag from http://wastetag.stanford.edu/ and attach to waste container as soon as first bit of waste is generated.
2. Clean all work surfaces after soldering. Treat all cleaning materials, including sponges or rags used to clean, as hazardous waste; do NOT dispose in regular trash streams.
3. Do NOT rinse sponges, rags, etc. in sinks.
4. Keep waste containers closed unless depositing materials into them.
5. Request pickup within 8 months of the first date of generation at https://wastepickup.stanford.edu.

For questions regarding soldering safety, contact Occupational Safety and Health (OSH) at (650) 723-0448, or at: healthandsafety@stanford.edu